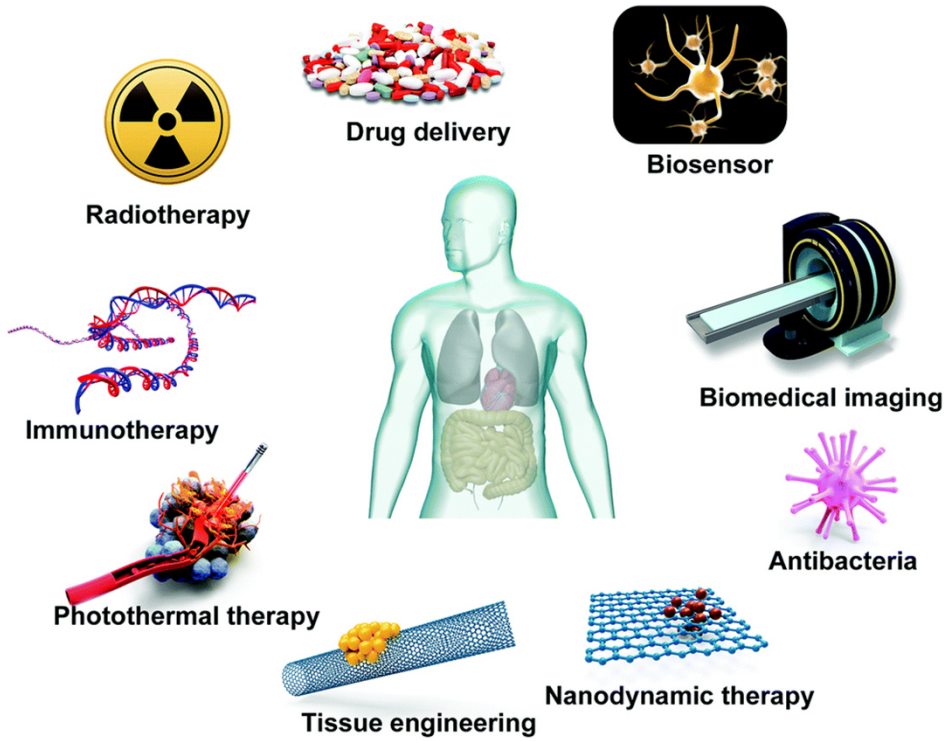
Introduction to Biomaterials

Lecture ZERO

An introduction to the course covering biomaterials fundamentals, curriculum, grading distribution, and main references

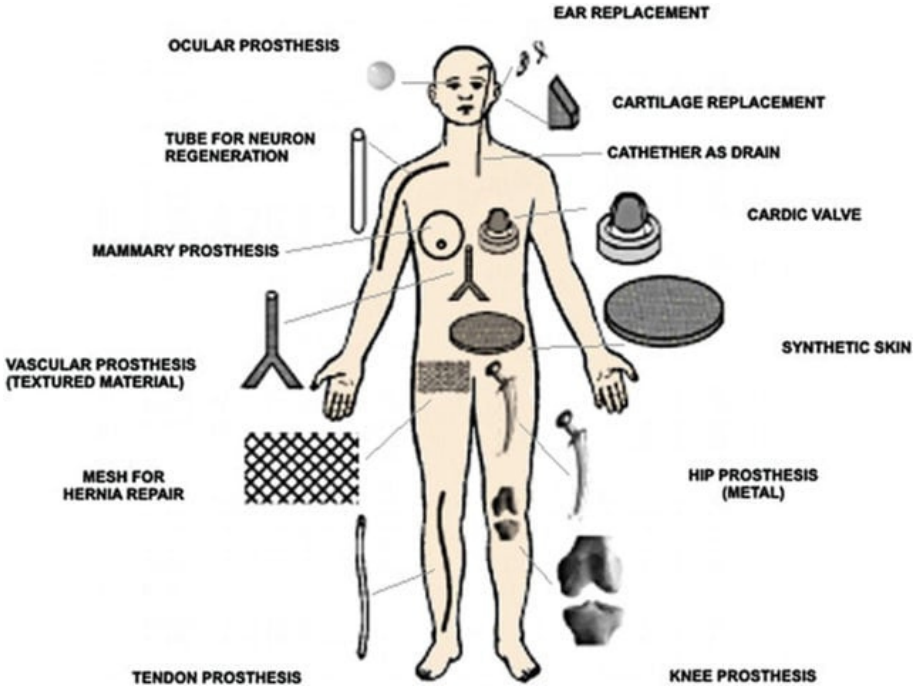
Introduction

- *WELCOME* everyone to the Biomaterials course.
- In this course, we will study how materials can be used inside the human body to repair, replace, or support biological functions.
- These materials are called biomaterials, and they play a key role in modern medicine.



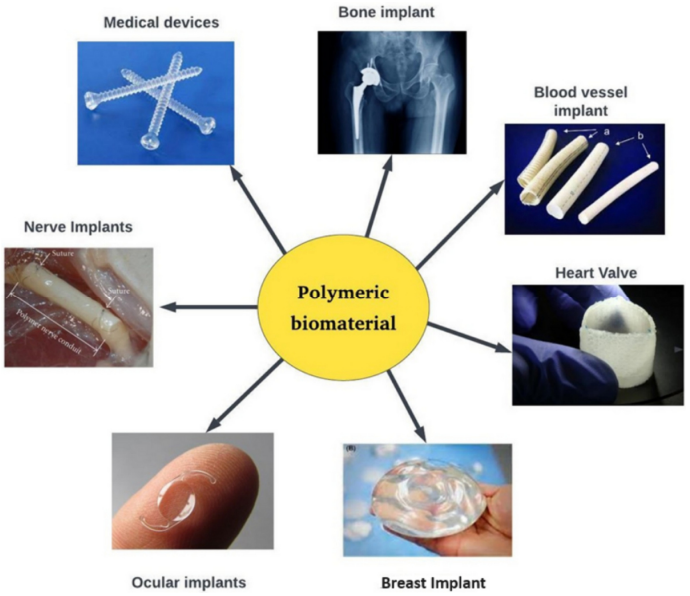
Introduction

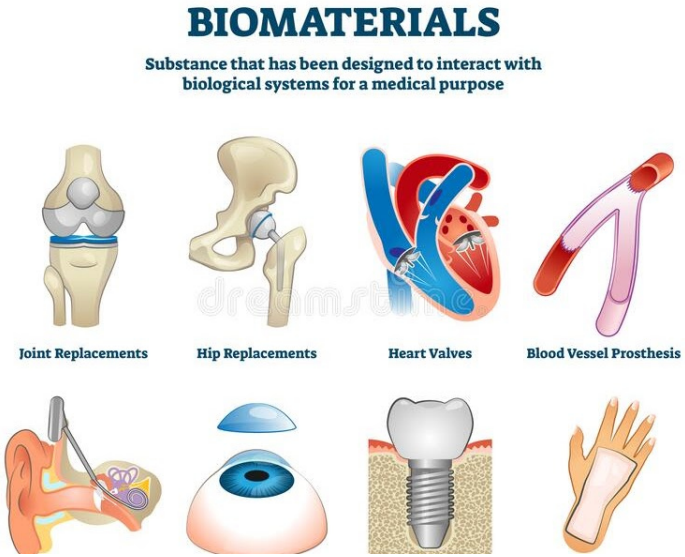
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Introduction

- You interact with biomaterials more than you think through implants, dental materials, medical devices, and drug delivery systems.





Why Biomaterials Matter ?

- Biomaterials have significantly improved the quality of life for millions of people worldwide.
- *They are used in:*
  - Implants (bone, dental, cardiovascular)
  - Drug delivery systems
  - Tissue engineering
  - Medical devices
- → Although biomaterials have existed for thousands of years, the field only became a recognized scientific discipline in the 20th century.

Example 1: Catgut Sutures



- Made from animal intestines (collagen)
- Used to close wounds
- Biodegradable (absorbed by the body)
- Early example of natural biomaterials



Example 2: Gold in Dentistry

- Used for fillings and crowns
- Biocompatible and corrosion-resistant
- Long-lasting inside the body
- Early example of metal biomaterials





Importance of the Course

This course is very important because biomaterials:

- Improve the quality of life for millions of patients
- Are essential in modern healthcare systems
- Are used in real applications like:
  - Artificial joints
  - Stents
  - Contact lenses
  - Tissue engineering
- Also, this field is growing very fast, which means there are strong opportunities for research, industry, and innovation.

Course Nature
(For Engineers, NO Pure Scientists)

- This course is designed mainly for **engineers**, not pure scientists.
- That means:
- We will focus on **applications and understanding**, not deep theoretical biology or chemistry
- The goal is to help you:
  - Understand how materials behave in the body
  - Choose the right material for a medical application
  - Think like an engineer solving a medical problem
- So don’t worry if your biology background is limited - we will cover what you need.

General Topics of the Course

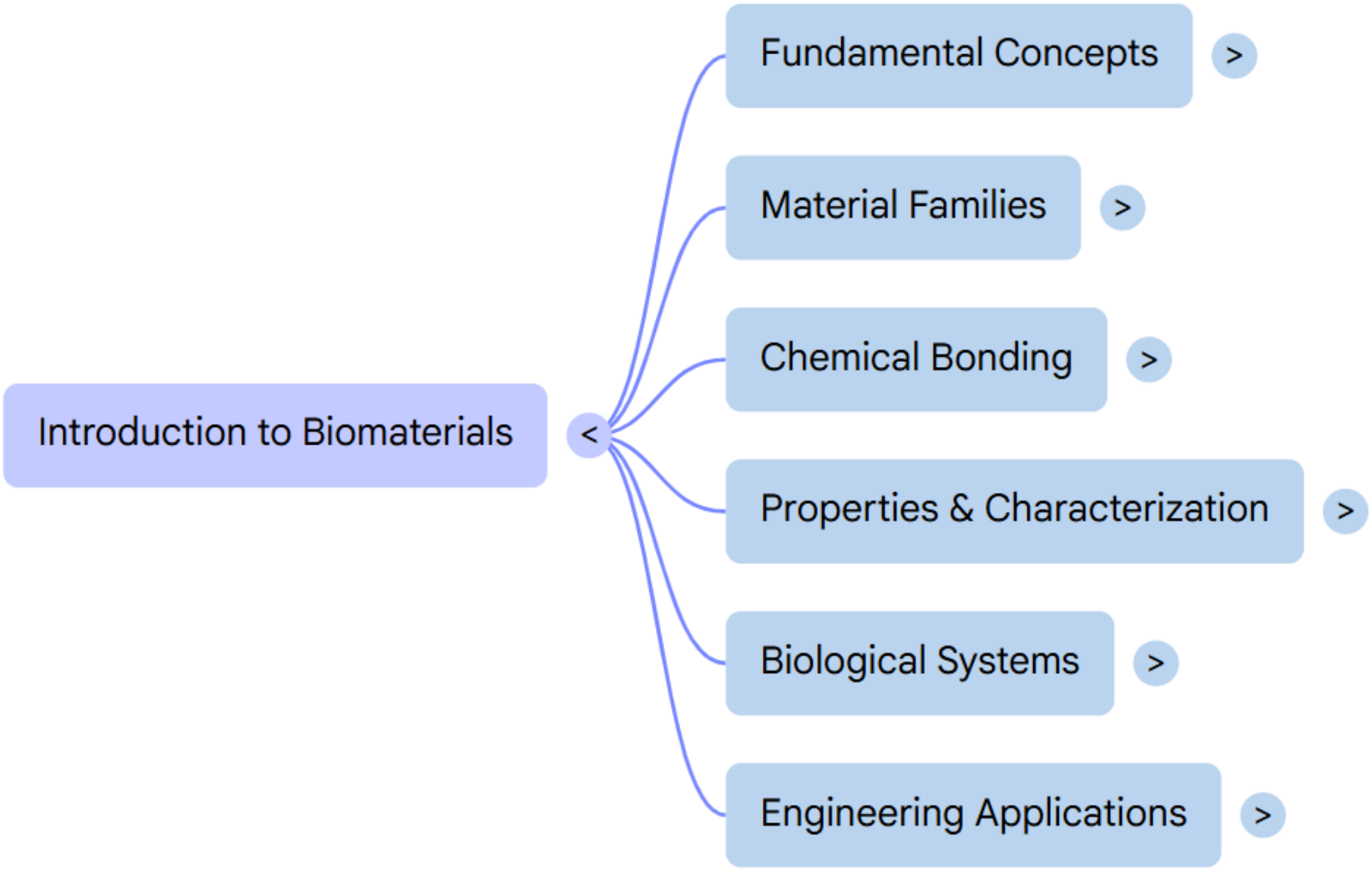
During this course, we will cover the main areas of biomaterials:

- Basics of materials science
- Basic biology concepts
- Types of biomaterials:
  - Metals
  - Polymers
  - Ceramics
  - Natural materials
- Surface modification and sterilization

General Topics of the Course

During this course, we will cover the main areas of biomaterials:

- Surface modification and sterilization
- How biomaterials interact with:
  - Proteins
  - Cells
  - Tissues
- Applications such as:
  - Drug delivery
  - Tissue engineering
  - Medical devices


Introduction to Biomaterials
Fundamental Concepts
Material Families
Chemical Bonding
Properties & Characterization
Biological Systems
Engineering Applications



Course Points Distribution

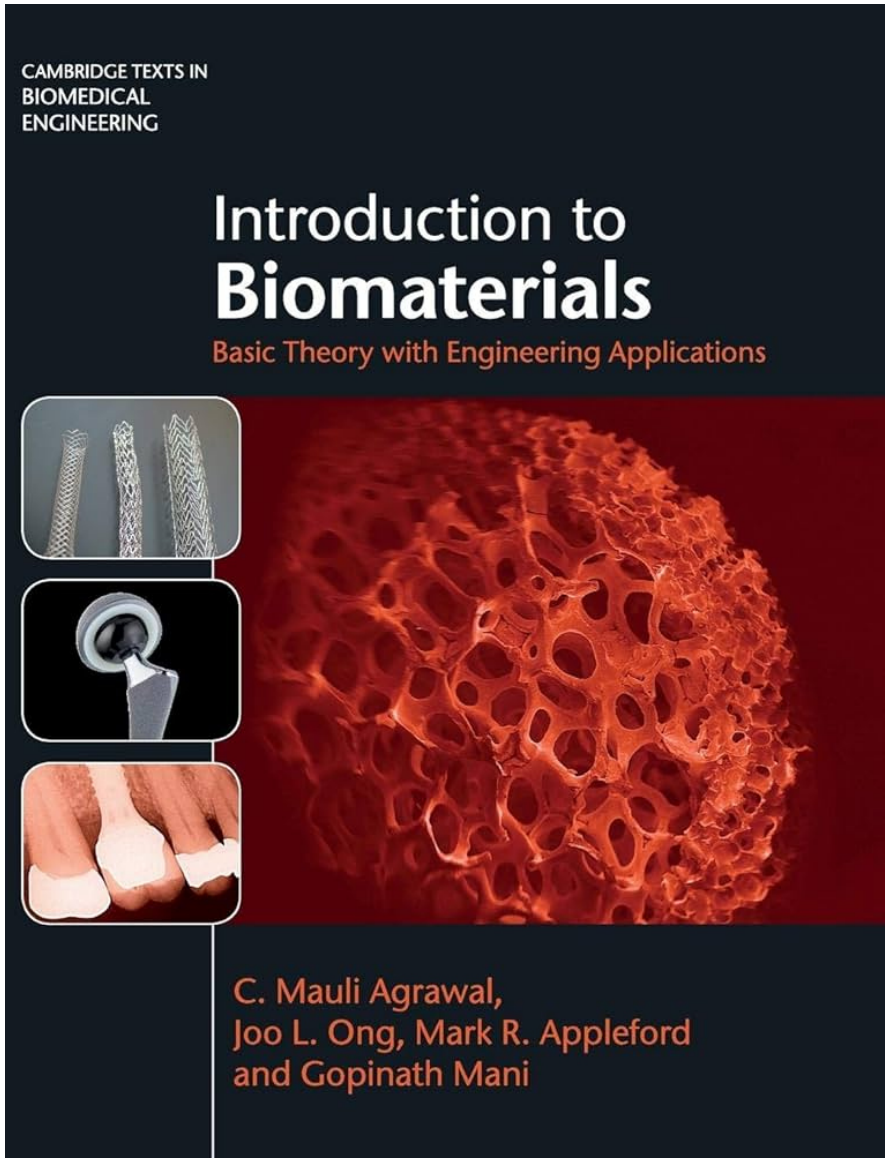
Assessment Method	Time	Marks
Homeworks / Assignments	-	20
Midterm Exam	1 Hour	20
Project	-	20
Final Exam	2 Hours	40
Total		100

Main Reference

- The main reference for this course is:

Introduction to Biomaterials: Basic Theory with Engineering Applications

- We will follow this book, but I will also provide:
  - Slides
  - Additional notes
  - Real examples



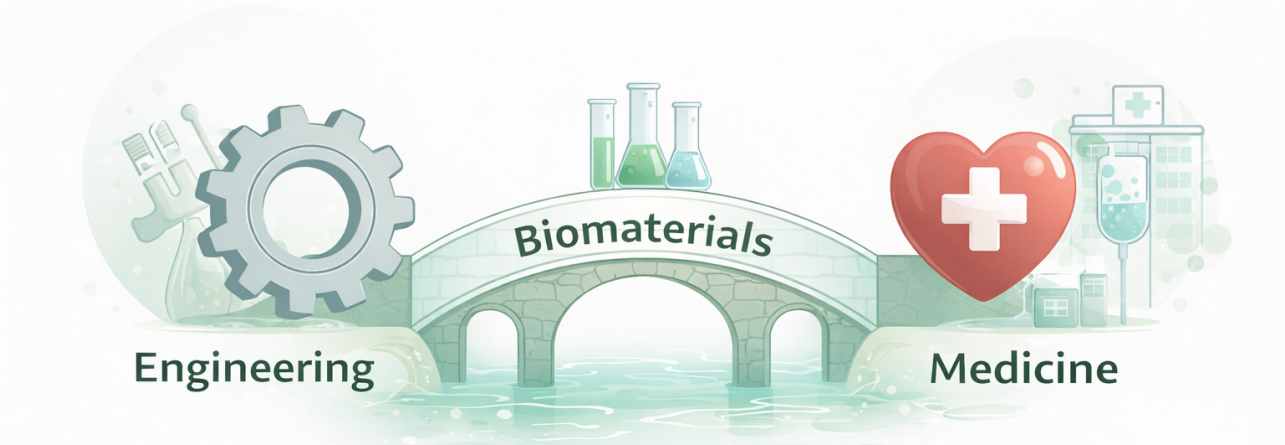
What We Will Focus On→

In this course, we will focus on:

- Understanding how materials behave in the body
- Linking engineering concepts to medical applications
- Developing your ability to:
  - Analyze problems
  - Choose materials
  - Think critically

Closing

- *By the end of this course*, you should have a clear understanding of biomaterials and be able to connect engineering knowledge with real medical applications.




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AL-AZHAR UNIVERSITY - GAZA

Thank you